

Mohammed Elmahfoudi

DevOps Engineer | Backend Developer

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Professional Summary

Results-driven Backend & DevOps Engineer with hands-on experience architecting scalable APIs, containerised deployment pipelines, and production-grade systems in healthcare and education. Skilled in CI/CD automation, RBAC security design, and full-stack development. Actively expanding expertise in infrastructure-as-code, cloud platforms, and distributed data engineering.

Experience

Backend Developer | SHBM INF Platform — UM6P

Jan 2025 – Present

[Next.js](#) · [Node.js](#) · [REST APIs](#) · [RBAC](#) · [PostgreSQL](#) · [Remote](#)

- Architected RESTful API services using Next.js API routes, powering student and administrative workflows for a university-scale platform
- Engineered a JWT-based RBAC system securing multi-tenant access across student, faculty, and admin roles; optimised DB queries and API response times for concurrent user loads

Full-Stack Developer | Independent / Startup Projects

2024 – Present

[NextJS](#) · [React](#) · [MySQL](#) · [Docker](#) · [GitHub Actions](#) · [Remote](#)

- Built and containerised full-stack applications with Docker Compose; implemented CI/CD pipelines via GitHub Actions enabling zero-downtime deploys and automated test runs on every push
- Developed Sahtek, a digital prescription platform connecting doctors and pharmacies, with an AI-powered drug alternative engine and audit-ready MySQL schemas for healthcare workflows

Entrepreneurship Intern | UM6P Explorer Program (MIT Sandbox)

Jun – Sep 2025

[Product Development](#) · [Market Validation](#) · [Agile](#) · [Benguerir](#)

- Validated a healthcare accessibility prototype with 20+ users; delivered working MVP under UM6P & MIT Sandbox mentorship using lean startup methodology to achieve confirmed product-market fit

Technical Projects

Sahtek — Digital Prescription Platform | [NextJS](#) · [MySQL](#) · [Docker](#) · [REST API](#) · [AI Integration](#)

- Designed microservice-ready backend for real-time prescription routing; integrated AI drug alternative module; full stack containerised with Docker Compose for one-command deployments

Ahlane — Community Health Platform | [React](#) · [Node.js](#) · [Prisma](#) · [WebSockets](#)

- Built real-time collaborative platform with WebSocket-driven live updates and Prisma ORM data layer on PostgreSQL supporting content filtering and user activity tracking

42 Network — Systems & Low-Level Engineering | [C](#) · [C++](#) · [Linux](#) · [POSIX](#) · [Sockets](#)

- Delivered peer-reviewed projects: custom Unix shell, HTTP server from scratch, memory allocator in C/C++; deep expertise in Linux processes, file descriptors, sockets, and POSIX APIs

Technical Skills

Languages	Python, JavaScript, TypeScript, C, C++, SQL, Bash
Backend	Node.js, NestJS, Express.js, Next.js API Routes, REST API Design
DevOps	Docker, Docker Compose, GitHub Actions, CI/CD Pipelines, Linux, Nginx, Bash Scripting
Databases	MySQL, PostgreSQL, MongoDB, Prisma ORM, Redis
Frontend	React.js, HTML5, CSS3
Data Eng.	ETL Pipelines, Data Modelling, Cloud Data Systems (ALX Africa)
Concepts	System Design, Microservices, RBAC/Auth, TCP/IP Networking, IaC (learning), Agile
Tools	Git, GitHub, Postman, Figma, VS Code, Docker Hub

Education

1337 Coding School (42 Network)	· Software Engineering Program	<i>2024 – Present</i>
Ecole Supérieure de Technologie, Kenitra	· DUT in Computer Science	<i>2022 – 2024</i>
ALX Africa	· Data Engineering Program (Remote) — ETL, data modelling, cloud architecture	<i>2025 – Present</i>